

Krishna Sai Rohith Vadla

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SUMMARY

Backend and AI engineer who ships LLM infrastructure that keeps running when providers go down. Built a production middleware system that eliminated provider outages as a point of failure across 12 models and 4 providers, an NLP pipeline compressing 35+ live sources at 93% reduction, and a client-facing dashboard running in production at a Wyndham-affiliated property. Strong in Python, FastAPI, and cloud deployment (AWS, GCP); published author at SPECOM 2023.

EDUCATION

Missouri University of Science and Technology

Master of Science, Computer and Information Sciences

Rolla, MO

May 2025

Indian Institute of Information Technology Dharwad

Bachelor of Technology, Computer Science and Engineering

Karnataka, India

May 2023

EXPERIENCE

Data Science Intern

Neoteksoft Inc.

Aug 2025–Present

Dallas, TX

- Built and deployed GPT-based text generation pipelines serving internal business workloads; containerized model serving with Docker and implemented CI/CD workflows, reducing deployment time from 3 hours to 15 minutes.
- Trained and deployed Transformer and CNN models using HuggingFace on AWS EC2 for classification tasks; migrated prototype pipeline from Railway/PostgreSQL to production AWS infrastructure with CloudWatch logging and automated model versioning.
- Engineered bilingual speech datasets for TTS model evaluation; conducted Mean Opinion Score testing with native speakers to validate production model quality before release.

Undergraduate Research Assistant

Indian Institute of Information Technology Dharwad

June 2022–May 2023

Karnataka, India

- Co-authored peer-reviewed research (SPECOM 2023, Springer LNAI 14339) on cross-lingual speech style transfer for Soliga, a zero-resource endangered tribal language; built a 10,000-sentence bilingual corpus from scratch and collected ~5,000 studio utterances per language with a custom recording UI.
- Improved synthesized speech intelligibility by 32% over the baseline model (MOS 3.02 → 3.99, evaluated by 20 native speakers across two language communities); reduced generator loss by 23% (32.1 → 24.7) via multi-scale reconstruction loss functions applied to VAE-GAN mel-spectrogram generation with WaveNet vocoder synthesis.

PROJECTS

LLM Request Router | *FastAPI, PostgreSQL* | Live Demo | GitHub

Dec 2025–Feb 2026

- Reduced failed request rate from 8% to under 1% during provider outages by building a REST API routing layer across 12 LLM models and 4 providers (OpenAI, Anthropic, Google, DeepSeek) with automatic failover, exponential backoff, and database-backed deduplication eliminating duplicate calls.
- Designed error classification logic distinguishing transient vs. permanent failures to prevent unnecessary retries; added request tracing and built a real-time dashboard displaying task status, retry counts, and provider health across 4 concurrent API sources.

Hotel Occupancy Dashboard (*Client Deployment*) | *Node.js, PostgreSQL* | Live Demo | GitHub

Aug 2025–Dec 2025

- Replaced manual paper-based occupancy tracking for a Wyndham-affiliated hotel with an automated dashboard, eliminating daily 2-hour commute for ownership to check property status; built automated ingestion pipeline parsing hourly SynXis PMS reports via Gmail IMAP with idempotent processing to prevent duplicate entries.
- Deployed dashboard used by ownership and general manager for real-time occupancy monitoring, pricing decisions, and staff scheduling; currently in production serving one property with signed agreement for expansion to two additional locations.

Personalized News Summarizer | *HuggingFace Transformers, SQLite* | Live Demo | GitHub

June 2025–Aug 2025

- Reduced article reading time by 93% (500–1000 words → 35–50 words) across 35+ RSS and NewsAPI sources by building an NLP pipeline using BART-based summarization and DistilBERT sentiment classification with 89% accuracy on test set.
- Added personalized recommendation engine using Sentence-BERT embeddings to rank articles by similarity to user reading history; deployed on Streamlit Cloud processing 200+ articles daily with sub-2-second page load.

OPEN SOURCE CONTRIBUTIONS

Boost Union | *Moodle Theme Plugin* | github.com/moodle-an-hochschulen/moodle-theme_boost_union

Apr 2026

- Contributed OAuth2 button color configuration feature to an actively maintained Moodle theme (5,000+ installs); implemented admin setting across `settings.php`, `lib.php`, and `lang/` files, passed PHPUnit and Behat test suites, resolved merge conflicts, and delivered production-ready code accepted by maintainers.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C/C++, SQL (PostgreSQL), HTML/CSS
Frameworks: FastAPI, React, Node.js, Flask, Streamlit
AI / ML: Hugging Face Transformers, PyTorch, scikit-learn, BART, DistilBERT, Sentence-BERT, OpenAI / Anthropic / Gemini APIs, multi-provider LLM routing, MLOps
Systems: REST APIs, microservices, distributed task queues, data pipelines, model deployment
Cloud & Tools: AWS (EC2, S3, CloudWatch), GCP, Docker, Git, Railway, MATLAB, VS Code
Libraries: NumPy, pandas, SQLAlchemy, SQLite, Recharts, Plotly, TanStack Query

CERTIFICATIONS

Generative AI Fundamentals — Databricks Academy	June 2025
Python Essentials 1 & 2 — Cisco	Sep 2025
Software Development Virtual Experience — Datacom	Apr 2026